The Cerebral Cortex And The Higher-order Cerebral Functions

Learning Objectives

1. Explain the anatomical organization of the cerebral cortex into neural layers and cortical columns
2. Discuss the concept of **primary, secondary and association cortical areas**
3. Describe the concept of lateralization and hemispheric dominance
4. Describe the location of the language centers and **identify typical language deficits**
6. Explain hemineglect syndrome and identify the cortical areas involved in attention
7. Tell the location and function of the main areas for visual perception and the deficits produced by damage of these areas
8. Explain the function of the prefrontal association areas and the main deficits seen after damage of these areas
9. Define the different types of memory and name the 2 main features of the amnesic syndrome

OBJ. # 1

The Cerebral Cortex

Neocortex - 6 layer cortex - 90% of all cortex in humans

Paleocortex - 3 layer cortex - Olfactory areas

Archicortex - 3 layer cortex - Hyppocampus

Each layer of the neocortex is composed of cells based on their sizes and shapes. They are classified either as PYRAMIDAL or non- pyramidal cells. Non- pyramidal cells are called stellate or GRANULE cells.

The Cerebral Cortex

OBJ. # 1

- **Layer I:** Molecular layer – Axons parallel to the surface of the cortex
- **Layer II:** External granular layer- Granule cells and small pyramidal cells – interlayer communication
- **Layer III:** External pyramidal layer – Mostly medium-sized pyramidal cells – horizontal communication
- **Layer IV:** Internal granular layer – granule cells – receives thalamo-cortical fibers
- **Layer V:** Internal pyramidal layer – large pyramidal cells – projection neurons to subcortical areas
- **Layer VI:** Multiform layer – Different types of neurons – origin of cortico-cortical projections and cortico-thalamic projections

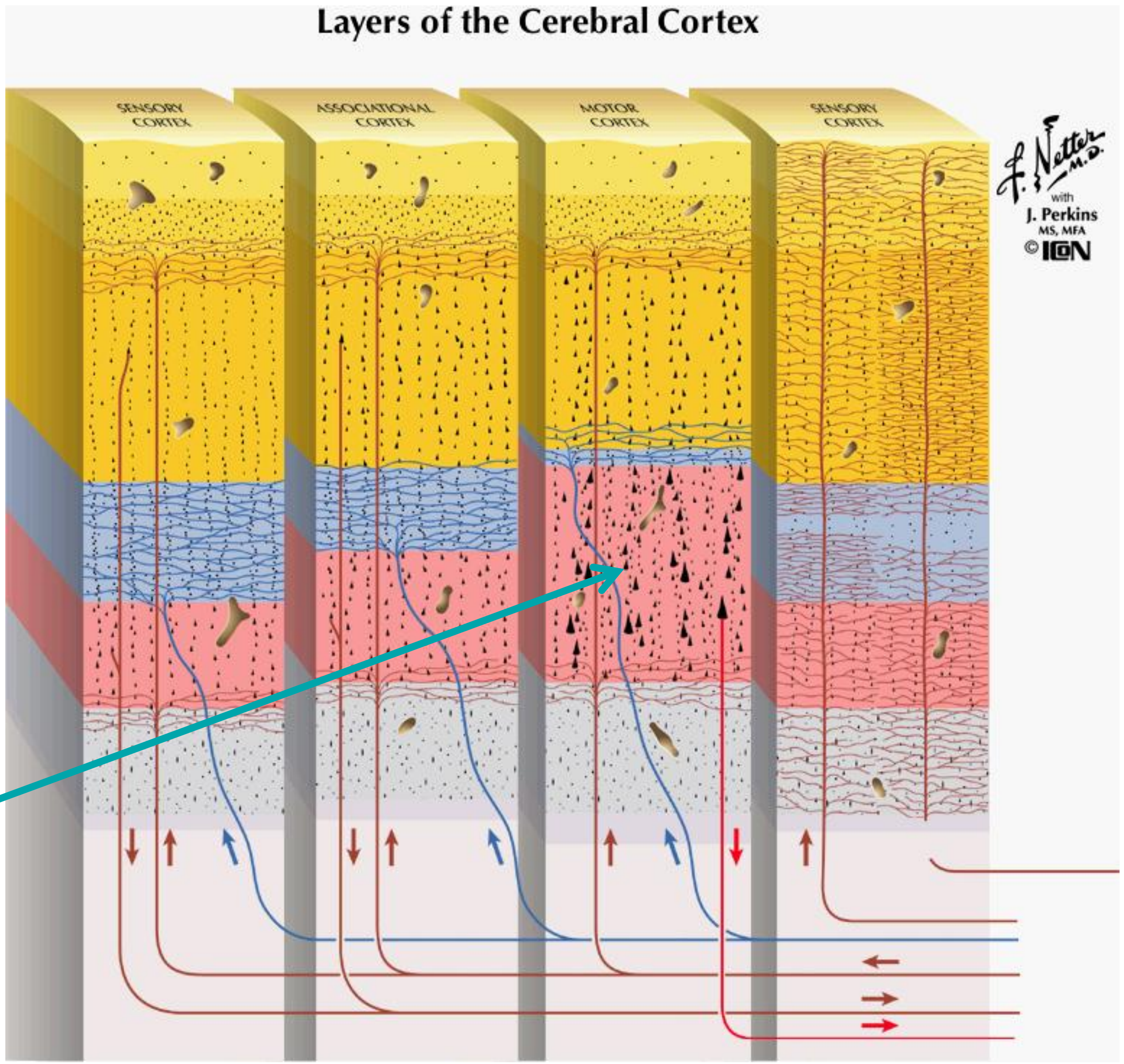
For your information only, no testable information

OBJ. # 1

The neocortex Layers and Connectivity

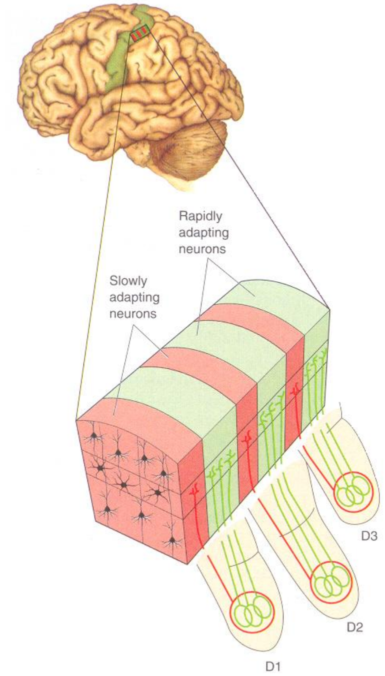
The neocortex forms ~90% of the cortical surface of the brain and has a laminated structure

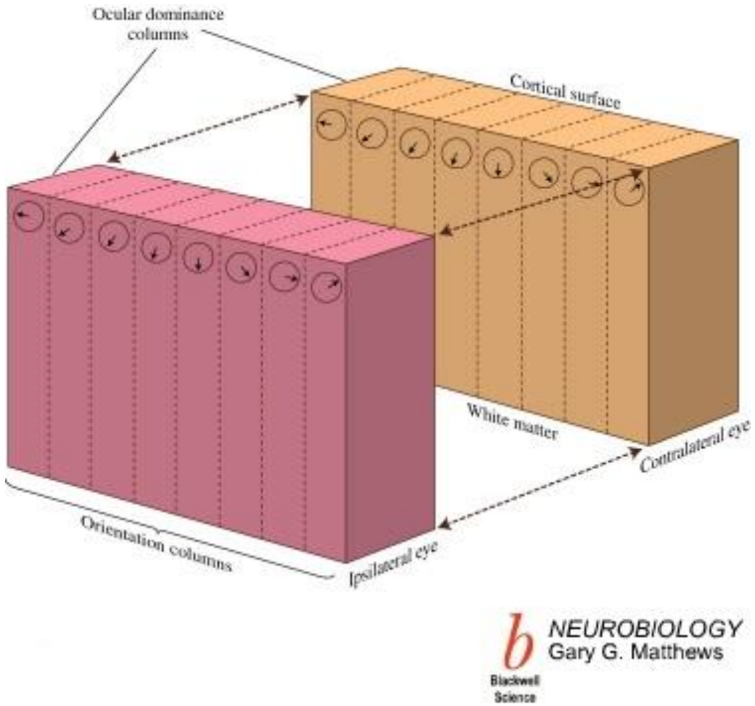
* Notice the relatively enlarged Layer V in the motor cortex, which is loaded with pyramidal cells projecting into the pyramidal tracts (corticospinal tracts)

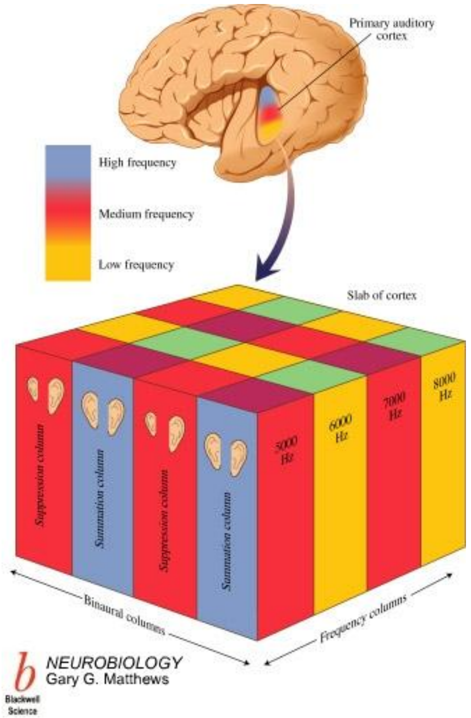


Cortical Columns

OBJ. # 1







In 1955 Vernon Mountcastle was the first to demonstrate the vertical, columnar organization of the cortex by physiological recordings in somatosensory cortices. An electrode traveling perpendicular to the pial surface encountered cells that responded all to the same modalities of sensation.

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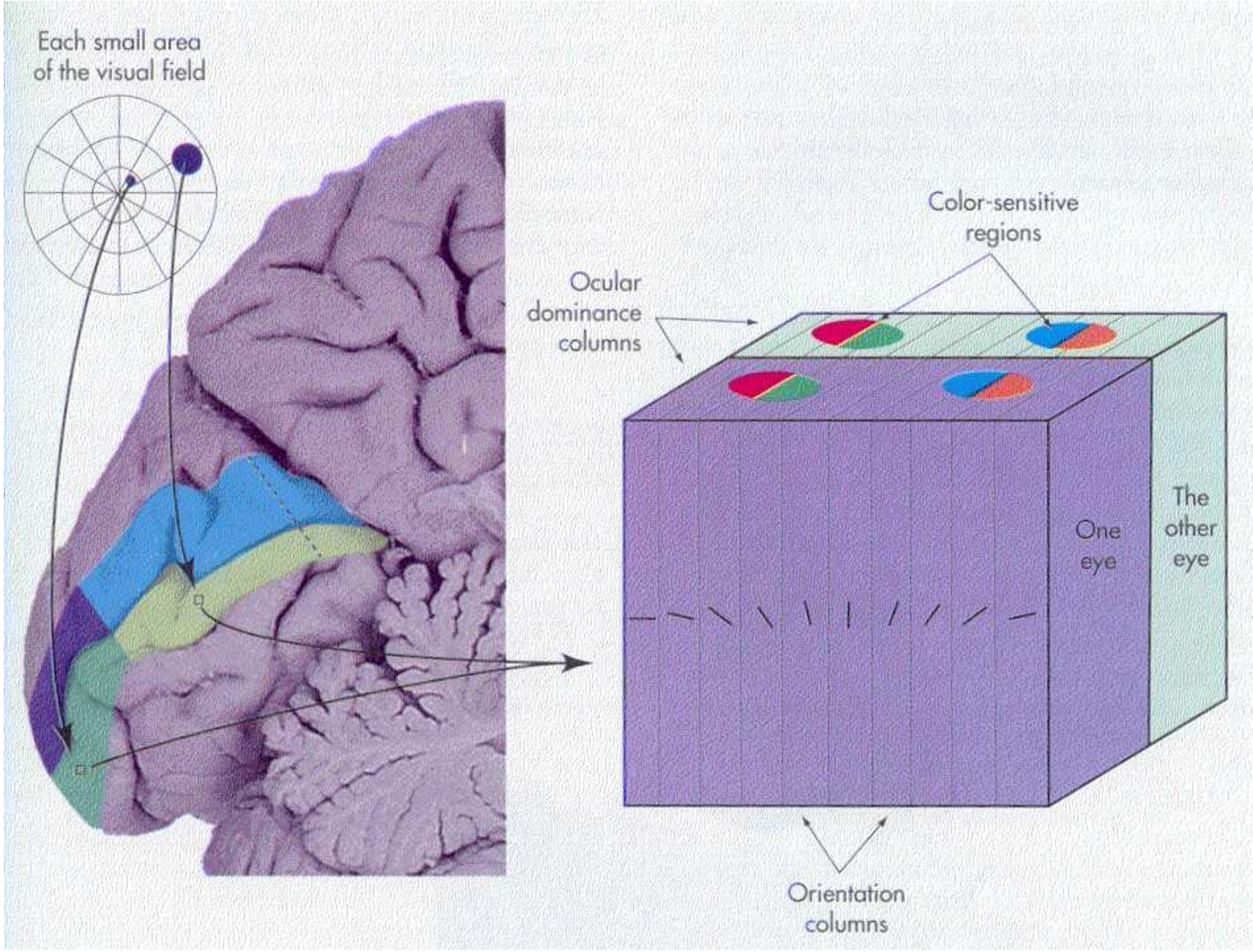
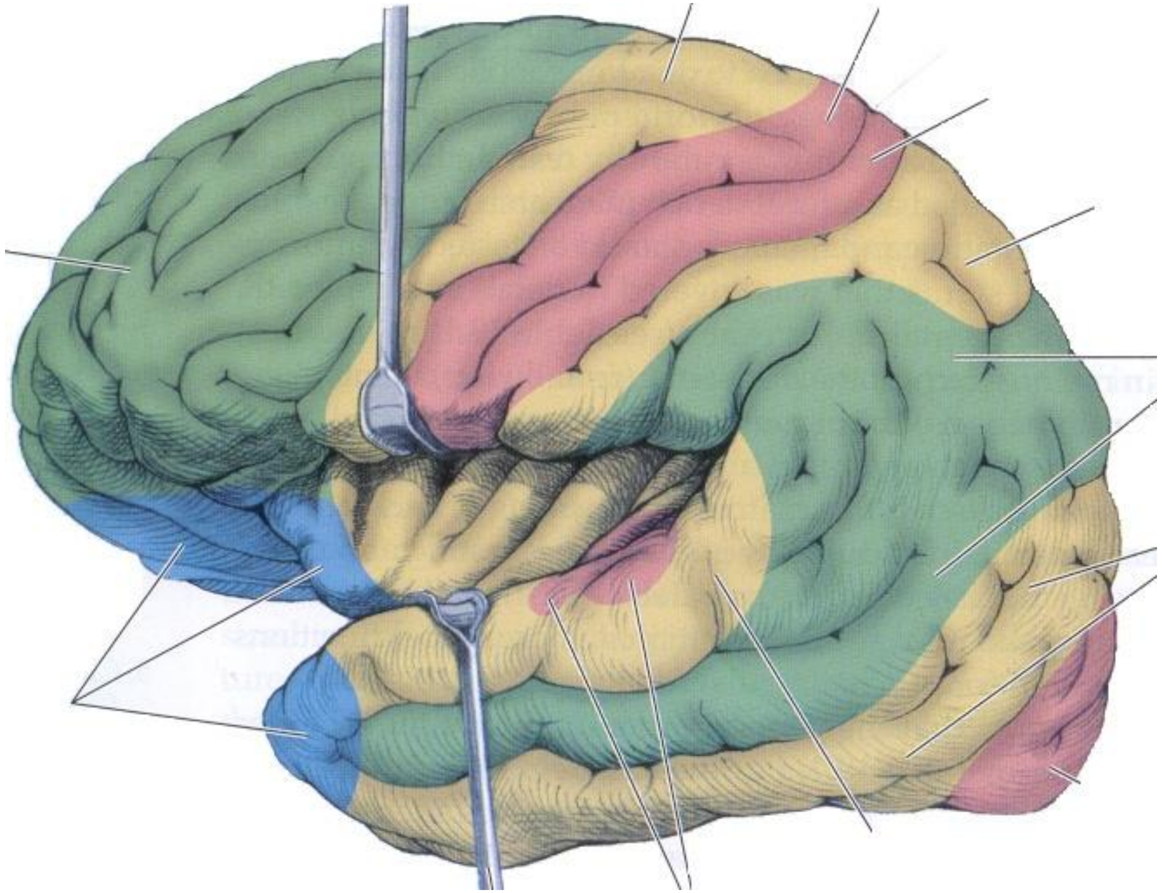


The Visual Cortex



Primary Cortices Are Unimodal

Unimodal Association

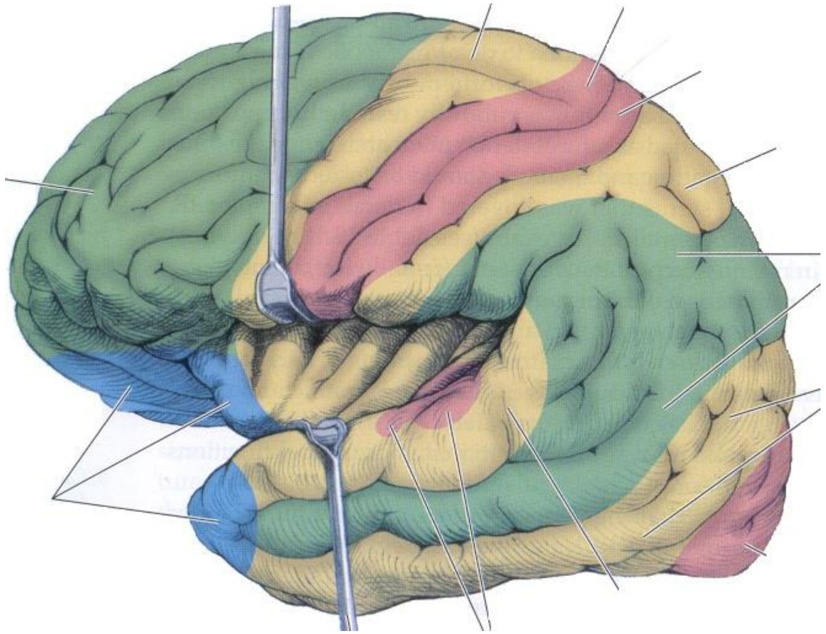
Multimodal Association

Primary Cortices

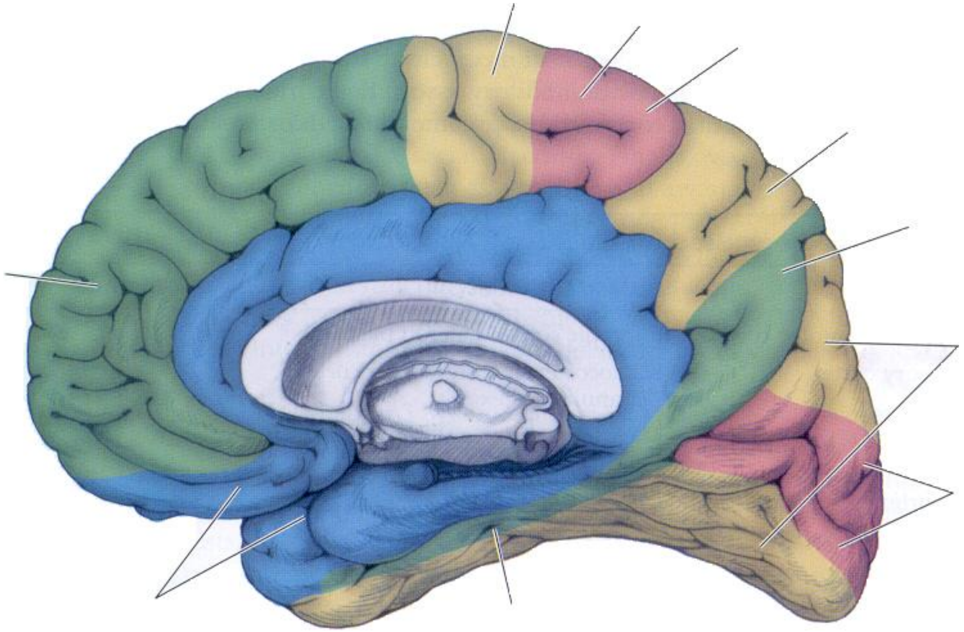
Limbic

OBJ. # 3

The Association Cortices



- There are secondary or unimodal association cortices and
- Multimodal or heteromodal association cortices



Yellow	Unimodal Association
Green	Multimodal Association
Pink	Primary Cortices
Blue	Limbic

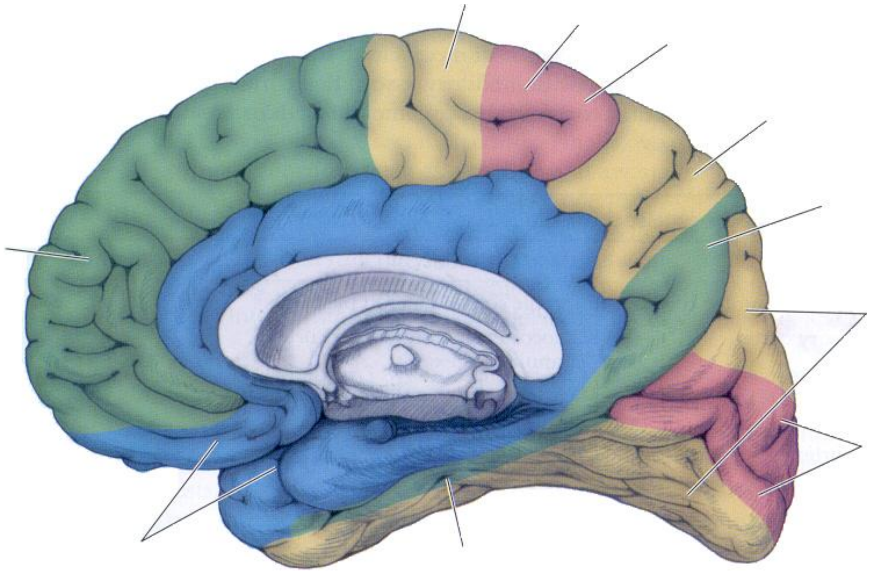
OBJ. # 3

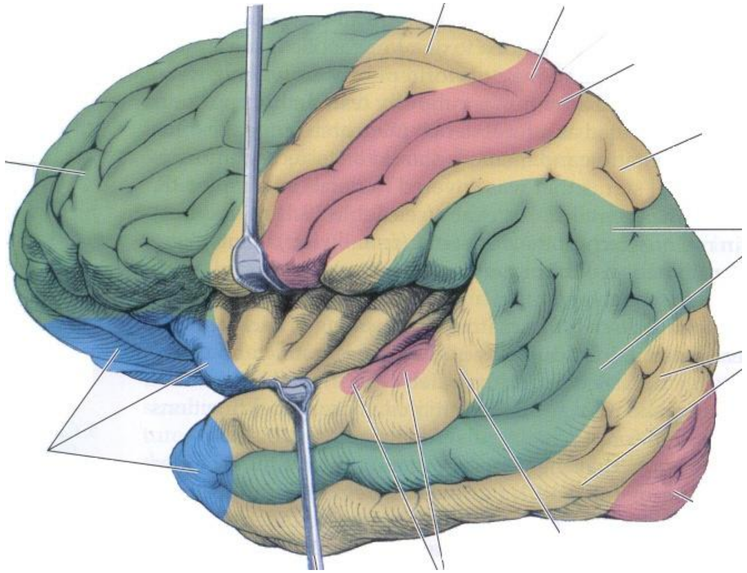
The Association Cortices

OBJ. # 3

There are 3 multimodal or heteromodal association cortices:

- Parieto-occipitotemporal association cortex
- Prefrontal association cortex
- Limbic association cortex





The Association Cortices

OBJ. # 3

The multimodal association cortices process more sophisticated aspects of several perceptions and are the site of the higher brain functions

A neat example of this is synesthesia, common in otherwise developmentally normal children:

SYNESTHESIA is a phenomenon by which people experience 2 or more perceptions as merged (syn= together and aesthesis=perception)

- Seeing blue when the note C sharp is played on the piano
- Seeing printed black numbers in colors: 3s are green, 5s are red!

http://www.youtube.com/watch?v=veoN1mh7RME

Hemispheric Lateralization

OBJ. # 4

- After experiments on split-brain patients, scientists understood that both cerebral hemispheres perform similar functions, they can independently perceive, learn and remember even though the processing mechanisms differ
- The left hemisphere is particularly good with analytical functions (language) while the right is better on spatial and musical perception.
- Both hemispheres are in constant communication via the **corpus callosum** and for many tasks they work together; other functions are totally lateralized.

Not to be confused with the 'left-brained,' 'right-brained,' pseudoscience personality stuff.

Hemispheric Lateralization

OBJ. # 4

Three typical examples of lateralization are

- Handedness
- Language
- Attention

TABLE 19.3 Functions of the Dominant and Nondominant Hemispheres

DOMINANT (USUALLY LEFT) HEMISPHERE FUNCTIONS	NONDOMINANT (USUALLY RIGHT) HEMISPHERE FUNCTIONS
Language	Prosody (emotion conveyed by tone of voice)
Skilled motor formulation (praxis)	Visual-spatial analysis and spatial attention
Arithmetic: sequential and analytical calculating skills	Arithmetic: ability to correctly line up columns of numbers on the page
Musical ability: sequential and analytical skills in trained musicians	Musical ability: in untrained musicians, and for complex musical pieces in trained musicians
Sense of direction: following a set of written directions in sequence	Sense of direction: finding one's way by overall sense of spatial orientation

OBJ. # 4

Handedness

- 90% of individuals are right-handed and process language in the left hemisphere.
- Two thirds of the left-handed people also process language in the left hemisphere and for the rest either the right side is dominant or both sides work together to control different language functions

- Note that left-handedness does not predict right lateralized language as well as right-handedness predicts left-lateralized language.
- Left-handedness is associated with *less lateralization*

OBJ. # 5

Language And Language Deficits

- Language is concerned with the use of a **system of arbitrary symbols** for the purpose of communication
- There is a set of rules for using these symbols that we call **grammar**
- The symbols must be ordered in a particular way that we call **syntax**
- We need to provide these symbols with appropriate emotional valence, we call this **prosody**

Aphasia is the partial or complete loss of the ability to produce or comprehend language after a lesion of the cerebral cortex. Most of the time this occurs without a loss of cognitive abilities or the loss of motor function to the muscles of speech.

OBJ. # 5

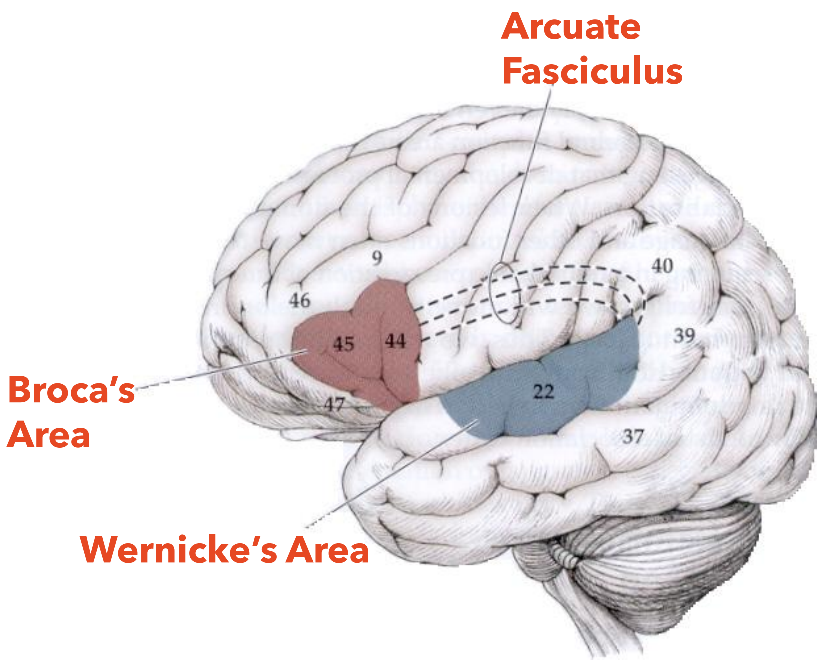
Aphasias

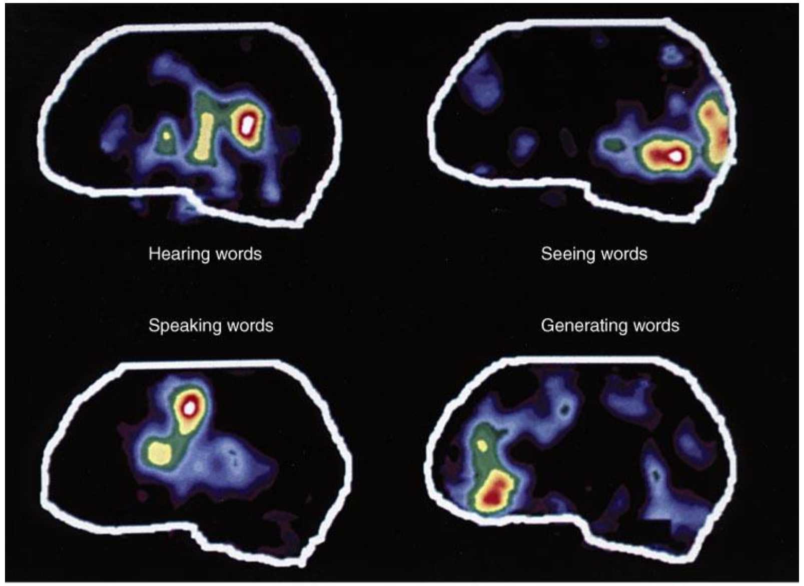
In 1864 the French Neurologist Paul Broca proposed that language is controlled by the frontal lobe of the left hemisphere in almost all cases. Today we call this area **Broca's area.**

In 1874 the German Neurologist Karl Wernicke proposed the existence of an area of the parietotemporal cortex also involved in the control of language. Today we call this area **Wernicke's area.**

These areas appear to be highly variable in different individuals, without clear anatomic delineation

OBJ. # 5





- Broca's aphasia
- Wernicke's aphasia
- Global aphasia
- Conduction aphasia

Two Major Brain Centers For Language With multiple interconnections

Classic Broca's Aphasia

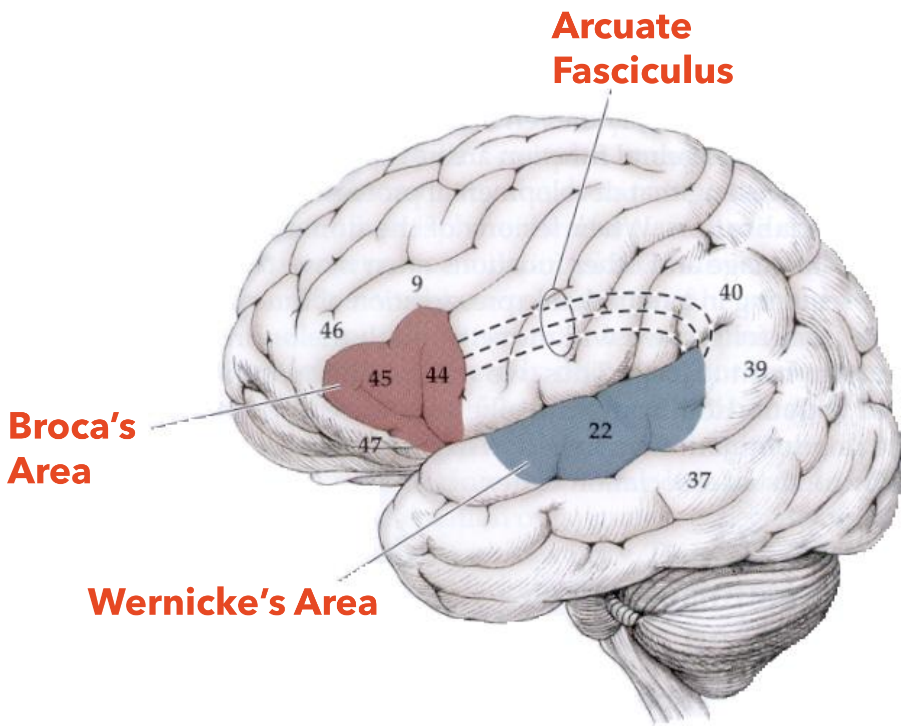
OBJ. # 5

Motor or Expressive aphasia: loss of the ability to produce language. The organizational aspects of language such as grammar and syntax are disrupted

http://www.youtube.com/watch?v=qYc6ddBanys

In these patients the speech is characterized by its loss of fluency. Patient have difficulty 'finding' the words they intend to speak, often stuttering or making errors. These paraphasic errors can be phonemic (swapping sounds,) or semantic (swapping meanings.)

Patient's become very frustrated by their inability to communicate



Classic Wernicke's Aphasia

OBJ. # 5

Fluent but "nonsense" speech with a marked loss of language comprehension in "heard" speech

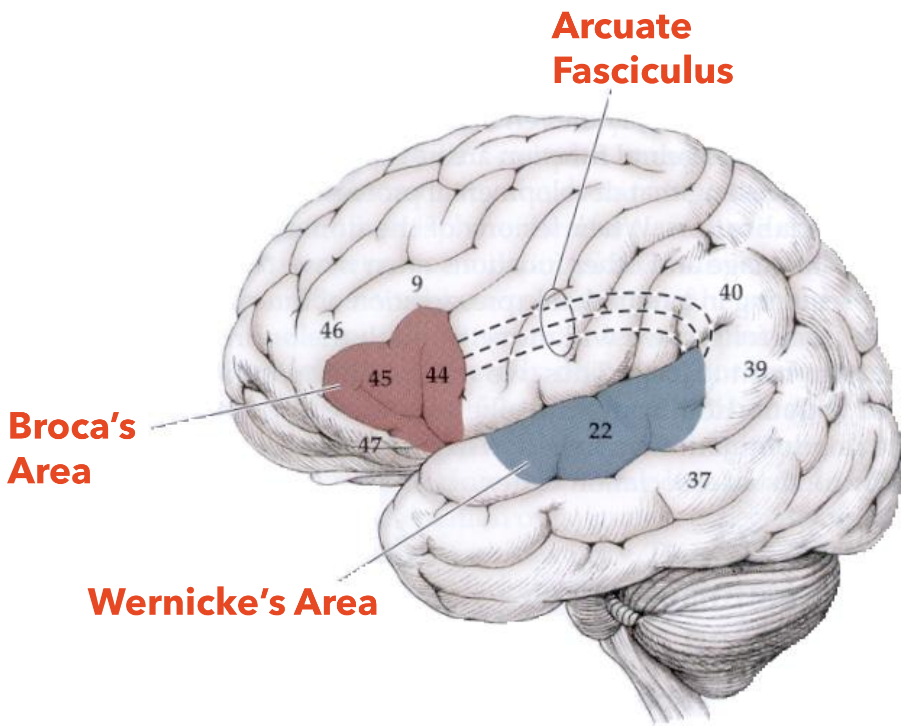
Grammar and syntax are adequate but the ability to appropriately link speech to meaning is lost – frequent use of "nonsense" words, paraphasic errors and words similar to the correct word. The patients have difficulty understanding simple commands.

Broca's

- Halting speech
- Difficulty with repetition
- Disordered syntax
- Disordered grammar
- Comprehension intact

Wernicke's

- Fluent speech
- Little repetition
- Syntax and grammar are adequate
- Impaired comprehension



Bedside Language Exam

OBJ. # 5

1. Spontaneous speech
   a. Informal interview
   b. Structured task
   c. Automatic sequences
2. Naming
3. Auditory comprehension
4. Repetition
5. Reading
   a. Reading aloud
   b. Reading comprehension
6. Writing
   a. Spontaneous sentences
   b. Writing to dictation
   c. Copying

1 – 4 will help you diagnose most language deficits with a quick, simple examination.

Aphasias

OBJ. # 5

Table 23-1

THE MAIN APHASIC SYNDROMES

TYPE OF APHASIA	SPEECH	COMPREHENSION	REPETITION	ASSOCIATED SIGNS	LOCALIZATION[a]
Broca's	Nonfluent, effortful, agrammatical, paucity of output but transmits ideas	Relatively preserved	Impaired	Right arm and face weakness	Frontal suprasylvian
Wernicke's	Fluent, voluble, well articulated but lacking meaning	Greatly impaired	None	Hemi- or quadrantanopia, no paresis	Temporal, infrasylvian including angular and supramarginal gyri
Conduction	Fluent	Relatively preserved	None	Usually none	Supramarginal gyrus or insula
Global	Scant, nonfluent	Very impaired	None	Hemiplegia usual	Large perisylvian or separate frontal and temporal
Transcortical motor	Nonfluent	Good	Largely preserved	Variable	Anterior or superior to Broca area
Transcortical sensory	Fluent	Impaired as Wernicke's	Largely preserved	Variable	Surrounding Wernicke area
Pure word deafness	Mildly paraphasic or normal	Impaired	Impaired	None or quadrantanopia	Bilateral (or left) middle of superior temporal gyrus
Pure word blindness (and alexia without agraphia)	Normal but unable to read aloud	Normal	Normal	Right hemianopia; unable to read own writing	Calcarine and white matter or callosum (or angular gyrus)
Pure word mutism (aphemia)	Mute, but able to write	Normal	None	None	Region of Broca area
Anomic aphasia	Isolated word-finding difficulty	Normal various sites	Normal	Variable	Deep temporal lobe

[a]Lesion in dominant (left) hemisphere unless noted.

Source: Adapted by permission from Damasio AR: Aphasia. *N Engl J Med* 326:531, 1992.

OBJ. # 5

Other Language Related Deficits

- **Alexia** – Reading impairment without sensory or motor deficits. This is an acquired reading deficit. Dyslexia is a developmental reading disorder
- **Agraphia** – Writing impairment without sensory or motor deficits. It is always present in patients with aphasia
- **Apraxia** – The inability to perform an action in response to a verbal command in the absence of any motor, comprehension, or coordination deficits
- **Gerstmann's Syndrome** – The lesion is in the angular gyrus of the parietal lobe. Patients present with:
  - Agraphia
  - Acalculia
  - Right – Left disorientation
  - Finger agnosia

OBJ. # 6

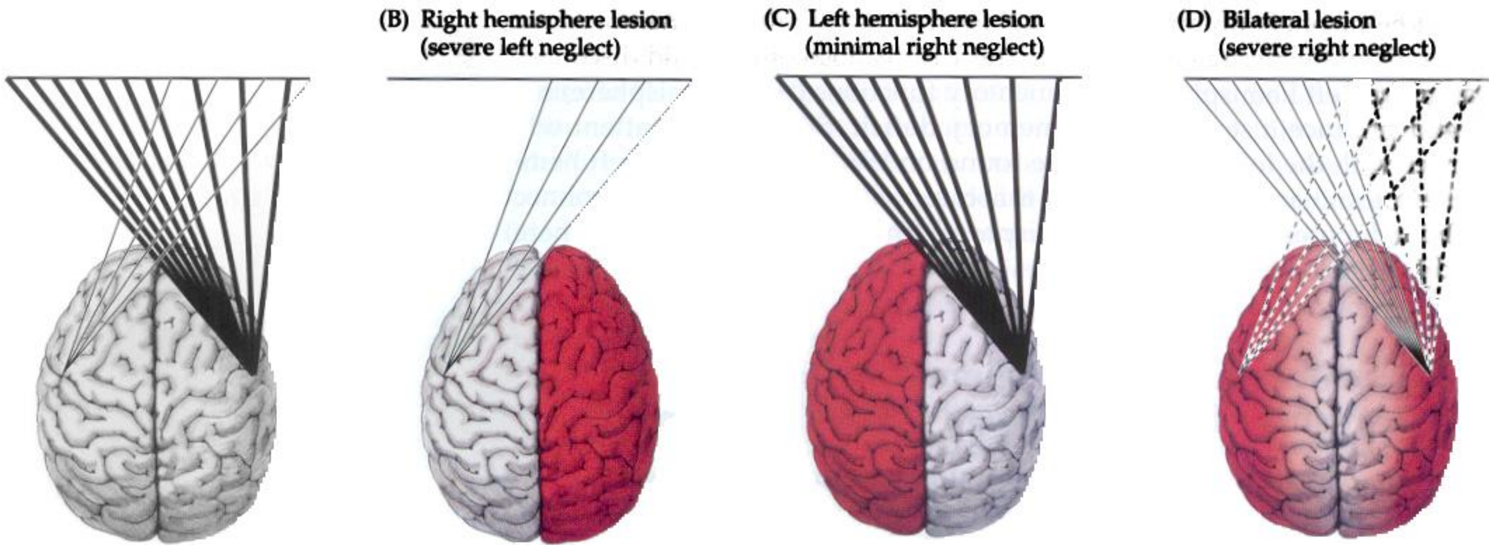
Attention And Spatial Processing

Attention is the ability of selectively process simultaneous sources of information coming to the brain and be able to separate one from the others

Attention depends on the activation of many different areas of the CNS. At cortical levels the posterior parietal, frontal and limbic areas of the right hemisphere are the most important for attention mechanisms

The parietal association areas of the non-dominant hemisphere are particularly important in the processing of spatiotemporal information and is becoming increasingly clear that the cortical eye movement systems are extremely important in directing attention

OBJ. # 6



NON-DOMINANT PARIETAL AREAS

- The left hemisphere responds to stimuli on the right space
- The right hemisphere respond to stimuli on both left and right spaces with more emphasis on the left
- Lesions of the right posterior parietal cortex produces
- Left Hemineglect Syndrome

Contralateral Neglect Syndromes

OBJ. # 6

There are 4 different aspects of the contralateral neglect syndrome

Sensory neglect – patients tend to ignore visual, tactile, or auditory stimuli on the contralateral side

Motor neglect – patients perform fewer movements with the contralateral limbs

Visual neglect – patients ignore contralateral visual inputs

Conceptual neglect – the patient exhibit lack of internal representation of the body or the external world contralaterally

ANOSOGNOSIA – Patients are not aware of the disease.

Contralateral Neglect Syndromes

OBJ. # 6

Drawing is a very useful test. The evolution of the patient can be observed by making the patient read, draw or describe a visual scene – in general patients read the right part of an article, or describe only the right portion of the scene, or draw the right portion of an object

These patients also exhibit personality and emotional changes with decrease levels of attention and alertness

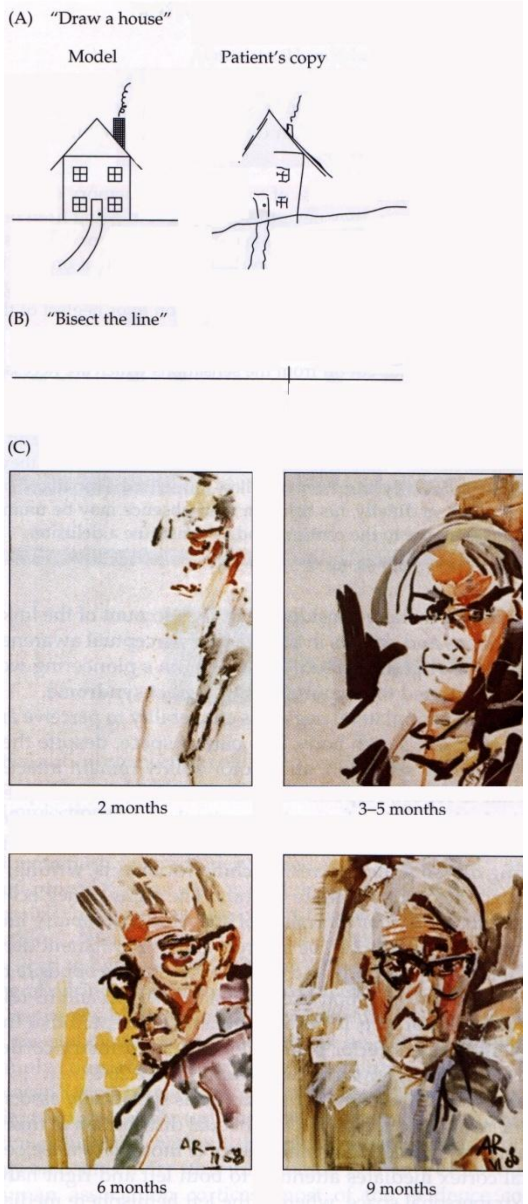
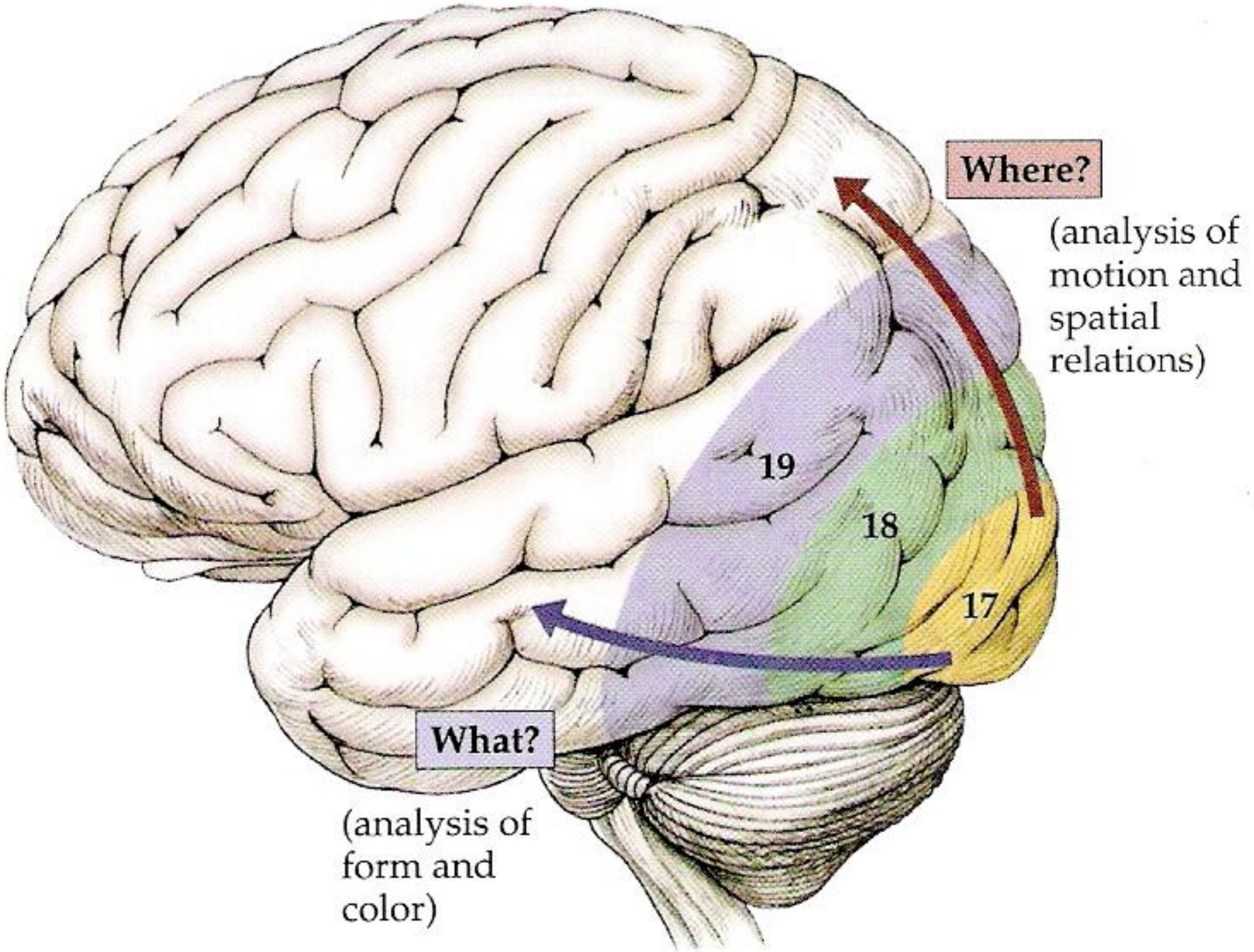


Figure 19.12 "What?" and "Where?" Streams of Visual Processing



Visual Association Cortices

OBJ. # 7

Cognitive Deficits Produced By Damage To Visual Cortical Areas

OBJ. # 7

Syndromes of the primary visual cortex

- If unilateral – contralateral homonymous hemianopia
  - Blindsight – Awareness of movement or light in the lost visual field (unilateral deficit)
- Cortical Blindness or Anton's Syndrome:
  - Produced by damage of the visual cortex bilaterally
    - Complete visual loss
    - Anosognosia
    - Lack of blink to threat

Cognitive Deficits Produced By Damage To Visual Cortical Areas

OBJ. # 7

Damage to the ventral pathway

Cortical color blindness or achromatopsia:

Produced by damage to area V4 on the ventral temporal lobe

Visual agnosia –

Produced by lesions of the ventral object area

Prosopagnosia –

Lesion of the face recognition area this deficit occurs mostly with bilateral lesions

Cognitive Deficits Produced By Damage To Visual Cortical Areas

OBJ. # 7

Damage to the dorsal pathway

Motion Blindness

Produced by lesion of the area MT and surrounding areas

Visual neglect

Produced by damage to the posterior parietal cortex

Balint’s syndrome

Produced by a bilateral lesion of the parieto-occipital cortex

Cognitive Deficits Produced By Damage To Visual Cortical Areas

OBJ. # 7

Damage to the dorsal pathway (posterior parietal)

Balint's syndrome

Patients have difficulty to scan a complex visual scene or identify moving objects. They can perceive small regions of the visual field at a time. Patients have optic ataxia, ocular apraxia and simultanagnosia

Optic Ataxia

Inability to reach an object in space under visual guidance or point to a target

Ocular Apraxia

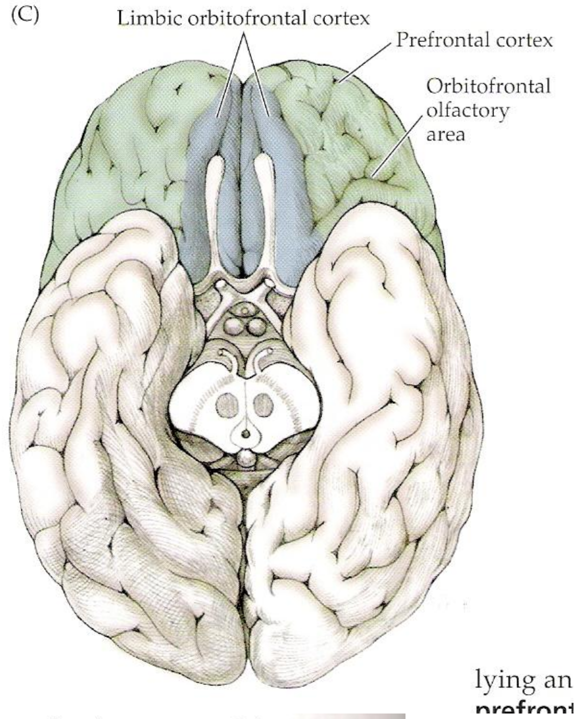
Difficulty in directing the gaze towards an object through saccades

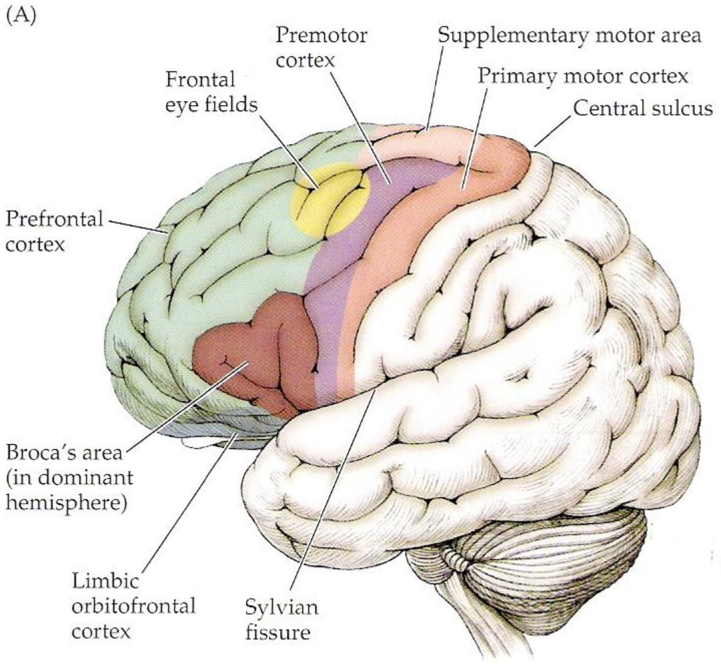
Simultanagnosia

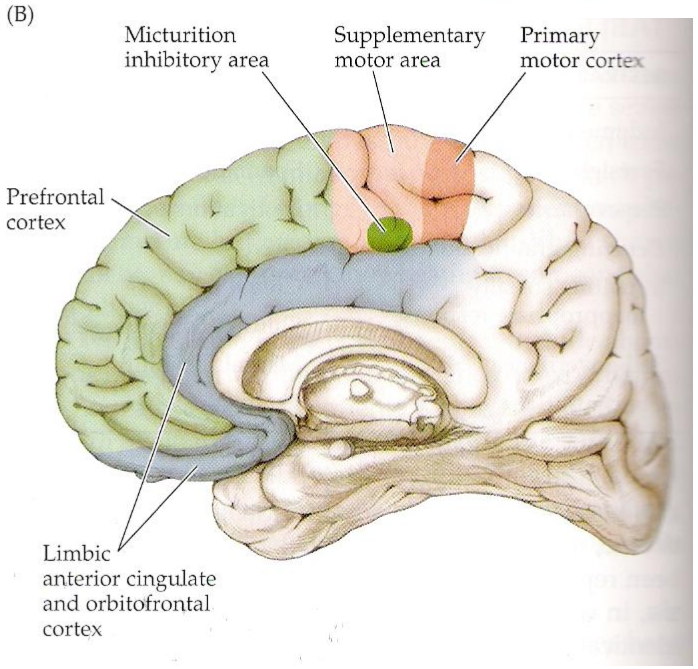
Inability to perceive more than one object in the visual field simultaneously

OBJ. # 8

Frontal Association Cortices The Prefrontal Cortex







Frontal Association Cortices - The Prefrontal Cortex

OBJ. # 8

- Frontal association areas are involved in planning and allow humans to function as effective and social beings.
- This region of the brain integrates information from all brain regions and is also implicated in the (maybe illusory?) concept of "self" - an individual's "personality".
- This brain area is also involved in short term memory storage and utilization.

Frontal Association Cortices The Prefrontal Cortex

OBJ. # 8

The prefrontal cortex oversees **executive function,** such as the control and sequencing of cognitive functions important in:

- Selection of the appropriate strategies to **solve problems**
- Motivation for productive and positive activities – **Initiative**
- The inhibition of incorrect responses mostly in relation to social behavior - **Restraint**
- The ability to deal with changes in focus (dealing with different things at the same time) and sequential tasks – **Order**

OBJ. # 8

Frontal Association Cortices The Prefrontal Cortex

- Being able to apply previously learned experience to new situations
- **Working memory** – Hold a limited amount of information for a short time while a number of cognitive operations are performed
- Integration of information from multimodal association cortices and limbic cortex in **decision making**

OBJ. # 8

Frontal Association Cortices The Prefrontal Cortex

Prefrontal cortex is defined as the areas of the frontal lobe anterior to area 6 and 8. Deficits of the prefrontal areas include:

- Deficits in goal-directed sustained mental activity, and the ability to change from one thought or action to another
- Akinesia and lack of initiative (apathy and abulia)
- Changes in personality particularly in mood and self control (disinhibition of behavior)

All these elements are frequently accompanied by frontal reflexes such as sucking, grasping and groping

OBJ. # 9

Memory Formation

Based on studies performed in humans and animals the main regions dedicated to memory are:

- **The medial temporal lobe** – includes the hippocampal formation and adjacent areas of the parahippocampal gyrus
- **The medial diencephalic memory areas** – includes: the medio-dorsal nucleus and anterior nucleus of the thalamus, the mammillary bodies and other nuclei lining the third ventricle
- It is important also to include the **white matter pathways that interconnect these areas** which are very important for memory consolidation and retrieval

Different forms of memory

OBJ. # 9

- **Declarative memory –** Also known as explicit memory – the conscious recollection of facts or experiences either bibliographical or on general knowledge
- **Nondeclarative memory** – Also known as implicit – the nonconscious learning of skills, habits and other acquired behaviors
- **Working memory** – The ability to hold a concept in memory long enough while doing a mental operation, such as an arithmetical operation
- **Recent memory** – The ability to recall lists of numbers or names for short periods of time – 4 to 5 minutes
- **Remote memory** – The ability to recall personal information or well-known events from the past

Different forms of memory

- **Sort-term memory -** Memories that last a few minutes
- **Long-term memory** – Memories that last longer than a few minutes

AMNESIA -

There are 2 features of the amnesic state:

- **Retrograde amnesia -** The impaired ability to recall facts or events that have been well established before the onset of the illness
- **Anterograde amnesia –** The impaired ability to acquire new information (learning)

OBJ. # 9